**BRAINWARE UNIVERSITY**
**SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES**
**DEPARTMENT OF COMPUTATIONAL SCIENCES**
**BACHELOR OF COMPUTER APPLICATIONS /**
**BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /**
**BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024**
**SEMESTER – IV**

**Course Code: BCA40204**
**Course Name: Data Warehousing and Data Mining**
**Weekly Contact Hours: 4L**
**Credits: 4**
**Total Allotted Hours: 60L**

**Course Objective:** The aim of the course is to understand the fundamental concepts and principles of data warehouse and Datamining, including its applications and the data mining process

**Pre-requisite(s):** Strong understanding of statistics, mathematics, programming, and database systems.

**Course Outcomes (COs):** After the completion of the course, students would be able to:

**CO1:** Define and understand the fundamental concepts and principles of data mining, including its applications and the data mining process.
**CO2:** Explain and apply various types of Datawarehouse architecture and Operations.
**CO3:** Illustrate and utilize a range of supervised and unsupervised learning algorithms to perform classification, regression, clustering, and association rule mining tasks.
**CO4:** Evaluate and validate data mining models using Clustering and Association rule techniques, and make informed decisions based on the results.
**CO5:** Apply advanced data mining techniques to formulate and analyse text data, social networks, time series data, and big data environments.

**Module 1: Introduction to Data Mining** **[10H]**

Overview of data mining and its applications, data mining process and methodologies, KDD, data pre-processing: data cleaning, integration, transformation, and reduction, Exploratory data analysis and visualization techniques.

**Module 2: Data Warehouse** **[12H]**

Introduction to Data Warehouse, differences between operational data base systems and data warehouse, data warehouse characteristics, data warehouse architecture and its components, extraction Transformation-Loading, Logical (Multidimensional), data modelling, schema design, star and snow-flake Schema, fact constellation, fact table, OLAP cube, OLAP operations, OLAP server architecture-ROLAP, MOLAP and HOLAP.

**Module 3: Classification and Regression** **[12H]**

Overview of classification and regression problems in data mining, decision tree algorithms (C4.5, CART), Bayesian classifiers (naive Bayes, Bayesian networks), Evaluation metrics for classification and regression models (accuracy, precision, recall, F1 score, mean squared error), Ensemble methods (bagging, boosting, random forests).

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**Module 4: Clustering and Association Rule Mining** **[13H]**

Clustering algorithms (k-means, hierarchical clustering, DBSCAN), Density-based clustering (OPTICS, Mean Shift), Association rule mining (Apriori algorithm, FP-growth algorithm), evaluation metrics for clustering and association rule mining (silhouette coefficient, lift, confidence, support).

**Module 5: Advanced Topics in Data Mining** **[13H]**

Text mining and sentiment analysis, Social network analysis and link analysis, time series analysis and forecasting, data mining in big data and distributed environments, ethical considerations and privacy issues in data mining.

**Course Outcomes (COs) and Modules Mapping:**

| Sl. No. | Course Outcome (COs) | Mapped Module(s) | Learning levels (as per Bloom's Taxonomy from K1 to K6) |
|---|---|---|---|
| **1.** | CO1 | M1 | K1, K2 |
| **2.** | CO2 | M2,M3, | K2,K3, K5 |
| **3.** | CO3 | M2,M3 | K2,K3 |
| **4.** | CO4 | M4 | K4, K5 |
| **5.** | CO5 | M5 | K3, K4, K6 |

***Text books:***

1. Data Mining and Data Warehousing, Parteek Bhatia, Cambridge University Press, 2nd ed., 2021.

***Reference books:***

1. Data Mining: Concepts and Techniques, Han, J., Kamber, M., & Pei, J., Morgan Kaufmann, 3rd ed., 2012.
2. Data Warehousing, Data Mining & OLAP , Alex Berson and Stephen J.Smith, Tata McGraw – Hill Edition, 35th Reprint, 2016.